

AERO2579 - Wind Tunnel Experiment 2

We have been learning in class about the growth of a boundary layer, now will be the chance to see the principle for yourselves. We will be using the same wind tunnel in lab 1, but this time we have a rig set up that will measure the velocity profile of the flow at certain distances from a flat plate.

REPORTING and SUBMISSION

This report is to be completed individually and submitted to canvas. Save the document as a PDF using the following naming convention:

FAMILYNAME_SXXXXXXX_Wind Tunnel Lab 2 AERO2579.pdf

This report is worth 20% of your final grade.

- *Ensure you change the header to your name and student number.*
- *Ensure that your work is presentable and easy to read, take care of formatting and text/graphic sizes. All graphs must be displayed digitally with the software of your choice.*
- *Marks will be deducted for poor presentation of work.*

LEARNING OUTCOMES

In this wind tunnel experiment, you will investigate the growth of a boundary layer over a flat plate, both for laminar and turbulent flow. The wind tunnel will be running at various speeds and it is required to interpret the data to determine the velocity profiles at the various locations along the flat plate. Using these profiles, the boundary layer thickness can be determined. It will then be required to plot the growth of the boundary layer over the flat plate.

It is highly recommended to record you wind tunnel lab and take plenty of photos to either use for evidence in your discussion section or to help remember how the test was conducted.

WIND TUNNEL

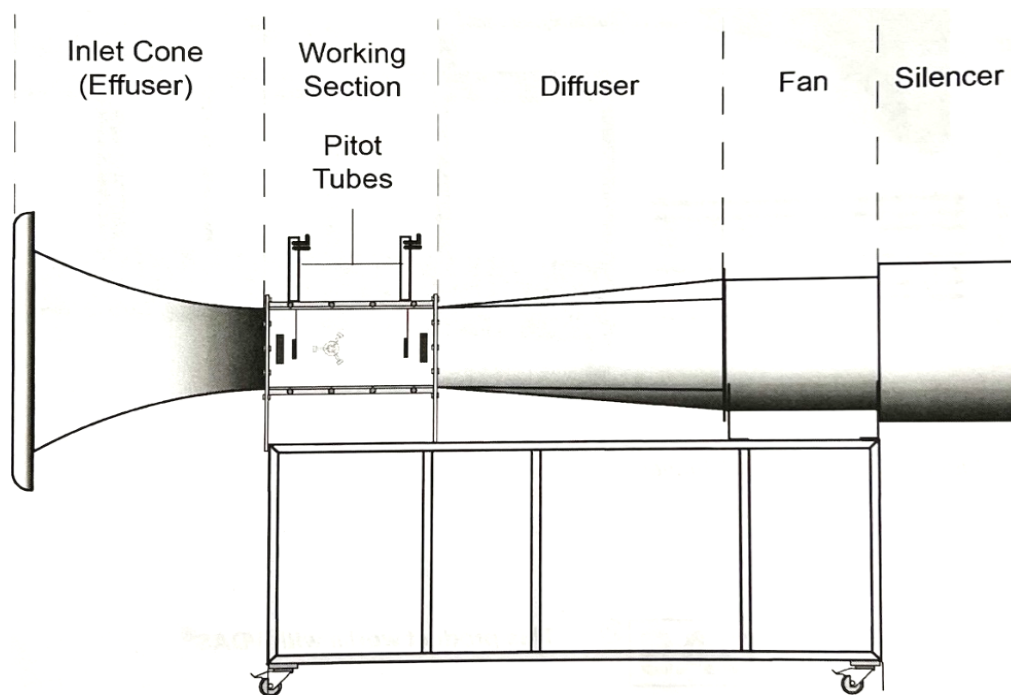


Figure 1: AF1300 General Layout

For this lab you will be using the AF1300 Subsonic Wind Tunnel in building 56 at RMIT. This wind tunnel provides a compact and practical open circuit wind tunnel in which the study of low-speed aerodynamic effects can be analysed. The wind tunnel can be seen in Figure 1 with the different sections identified.

Air enters the tunnel through an aerodynamically designed diffuser (cone) that accelerates the air in a linear manner. It then passes through a honeycomb flow straightener and enters the working section, then passing through a grille before moving through a diffuser and then to the variable speed axial fan. The grille protects the fan from damage by loose objects. The air leaves the fan, passes through a silencer unit and then back out to atmosphere.

The speed of the axial fan (and therefore the air velocity in the working section) is controlled by an electronic drive control in the separate Control and Instrumentation unit. The Control and Instrumentation unit houses facilities for many of the optional ancillaries.

Flat Plate Boundary Layer Model

The AF1300f Flat Plate Boundary Layer Model works with the AF1300 Subsonic Wind Tunnel. It shows boundary layer development and separation.

The Flat Plate Boundary Layer Model is just one of a range of models that TecQuipment manufacture to accompany our range of wind tunnels. Each model allows a wide variety of experiments and demonstrations to be conducted.

The Flat Plate Boundary Layer model is made of a hinged stainless steel plate. When fitted inside the working section of a wind tunnel, the angle of the plates can be adjusted to set the optimum conditions for the experiment. Fine filament tufts are provided to aid visualisation of the flow over the plate.

On the upper surface of the model are five small aerofoils set at right angles to the surface. Each aerofoil is drilled with five tiny pitot holes on the leading edge. Each hole is connected to a separate tube (25 tubes in total). All the tubes are routed together and emerge at the side of the model. The aerofoils are staggered so that their wakes do not interfere with each other.

Due to the static pressure being constant through the height of the boundary layer, only a single static pressure is required to compare against the pitot pressures to calculate the velocity at each pitot location. The static pressure is measured off the model off the wall of the test section, wall pressure.

When the model is fitted in the wind tunnel, the tubes may be connected to the AFA 1 Multi-tube manometer (or the AFA6 32-way Pressure System) for pressure measurements.

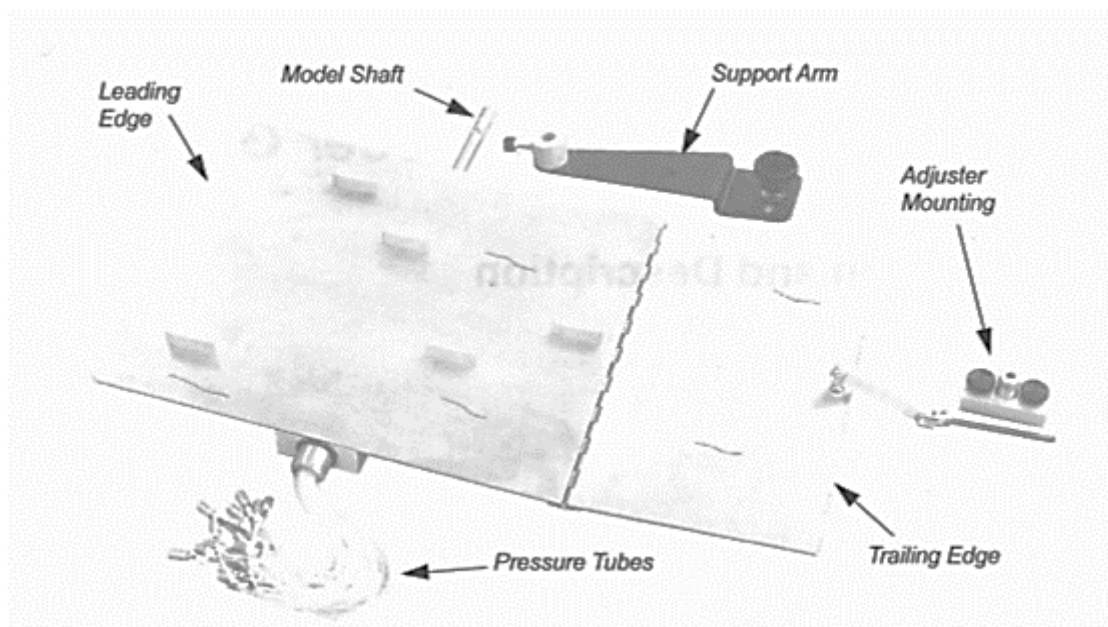


Figure 2: The Main Parts of the Flat Plate Boundary Layer Model

Boundary Layer

Technical Details

Item	Details
Overall Dimensions	Hinged plate 300 mm x 526 mm long
Nett Weight	5 kg
Tappings	5 x 5 groups = 25
Tapping Positions	See Figure 3
Tappings Heights	See Figure 4

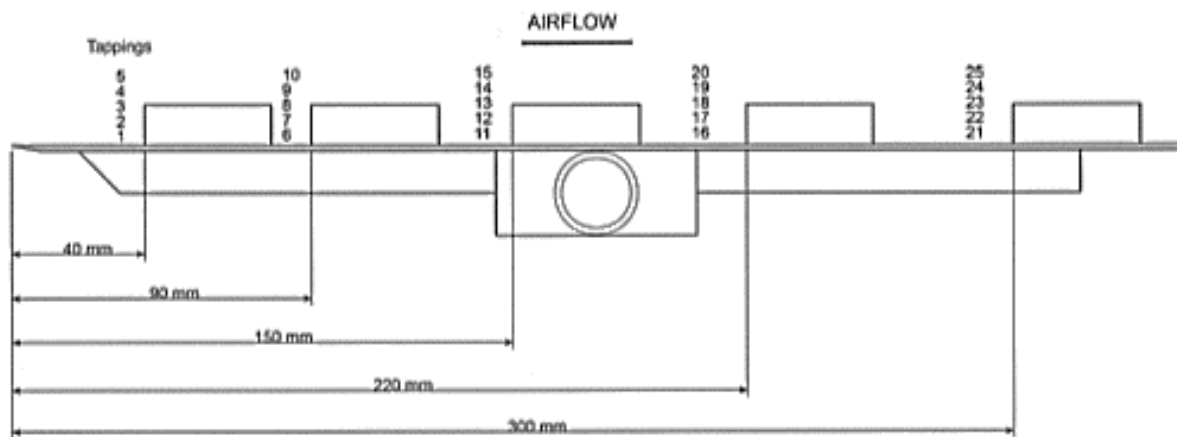


Figure 3: Horizontal Distances from the Leading Edge of Each Aerofoil

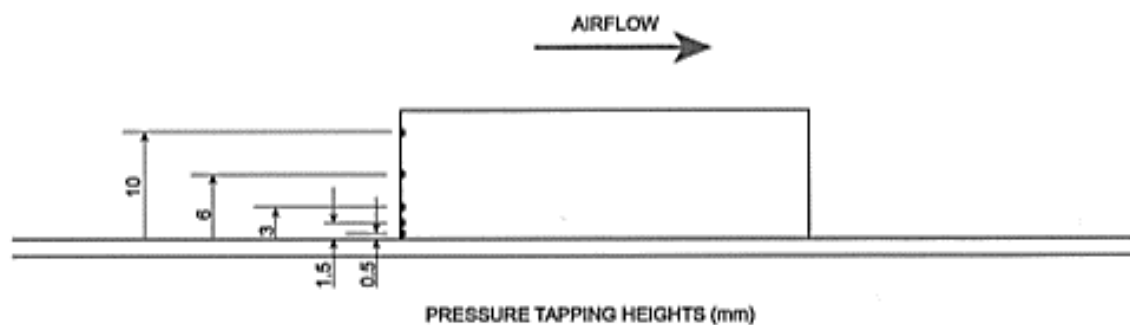


Figure 4: Vertical Distances of Pitot Holes from Baseplate

Acronyms

Symbol	Meaning	Units
h_T	Height of water in manometer due to model tapping pressure	mm _{H2O}
h_w	Height of water in manometer due to static wall tapping pressure	mm _{H2O}
η	Kinematic viscosity of air	m^2/s
μ	Dynamic viscosity of air (at 20°C)	$1.8 \times 10^{-5} \text{ kg/m}$
P_a	Ambient pressure	Pa
P_T	Model tapping pressure	Pa
P_w	Static wall tapping pressure	Pa
R	Gas constant	$287 \text{ J kg}^{-1} \text{ K}^{-1}$ for dry air
T_a	Ambient temperature	K
u	Local velocity	m/s or $m.s^{-1}$
u_0	Freestream velocity	m/s or $m.s^{-1}$
x	A horizontal position (of a set of pressure holes)	m (or mm where shown)
y	A vertical position (of a set of pressure holes)	m (or mm where shown)
δ	Boundary layer thickness	m
ρ	Air density	kg/m^3 or kg.m^{-3}

Conversions

1 mm H₂O at 20°C = 9.806 Pa

0°C = 273 K

Air Density

The density of the air in the wind tunnel can be found from a ratio of the ambient pressure, the gas constant and the ambient temperature:

$$\rho = \frac{P_a}{RT_a} \quad (1)$$

Pressure (P) and Velocities (u) and (u_0)

The static pressure P_s/P_w in the wind tunnel is that found by the wall tapping in the side of the working section, or the static pressure holes in the side of a pitot-static probe.

The local pressure P_T is that found at any point or tapping in the airflow of the wind tunnel

A simplified version of Bernoulli's equation gives,

$$P_T = P_w + \frac{1}{2}\rho u^2 \quad (2)$$

from which the **local velocity** at any point in the wind tunnel can be found from:

$$u = \sqrt{2 \times \frac{P_T - P_w}{\rho}} \quad (3)$$

For accurate comparison with theory and a datum point, it is necessary to know the **free stream velocity** u_0 , which is the same as the wind tunnel reference velocity. From this the velocity ratio can be created:

$$\frac{u}{u_0} \quad (4)$$

Reynolds Number (Re)

During the 19th century, an Irish physicist, Osborne Reynolds experimented with fluid flow. His experiments led him to create an equation that helps to simplify the variables in many equations of fluid dynamics:

$$Re = \frac{\rho u L}{\mu} \quad (5)$$

or as shown in some textbooks:

$$Re = \frac{uL}{\nu} \text{ as } \nu = \mu/\rho \quad (6)$$

In terms of fluid flow along a flat plate with surface length L :

- A Reynolds number greater than 100,000 – 500,000 indicates turbulent flow.
- A Reynolds number less than 100,000 indicates laminar flow.

So, in terms of the flow along a flat plate, for any position x :

$$Re_x = \frac{\rho u_0 x}{\mu} \quad (7)$$

Where x is any location from the leading edge to the point of interest.

The nature of the process of transition renders it prone to factors such as turbulence in the free stream and surface roughness of the boundary, so it is impossible to give a single value of Re_x at which transition will occur, but it is usually found in the range 100000 to 500000.

Velocity in the Boundary Layer

According to theory, where a boundary layer exists, the velocity of the air moving along the surface is slowest at the surface and gradually increases in an asymptotic manner as it moves away from the surface and into the free stream. See Figure 5

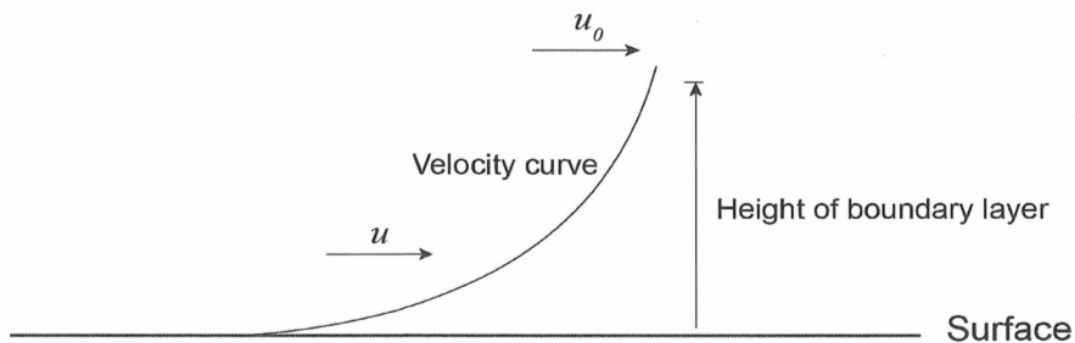
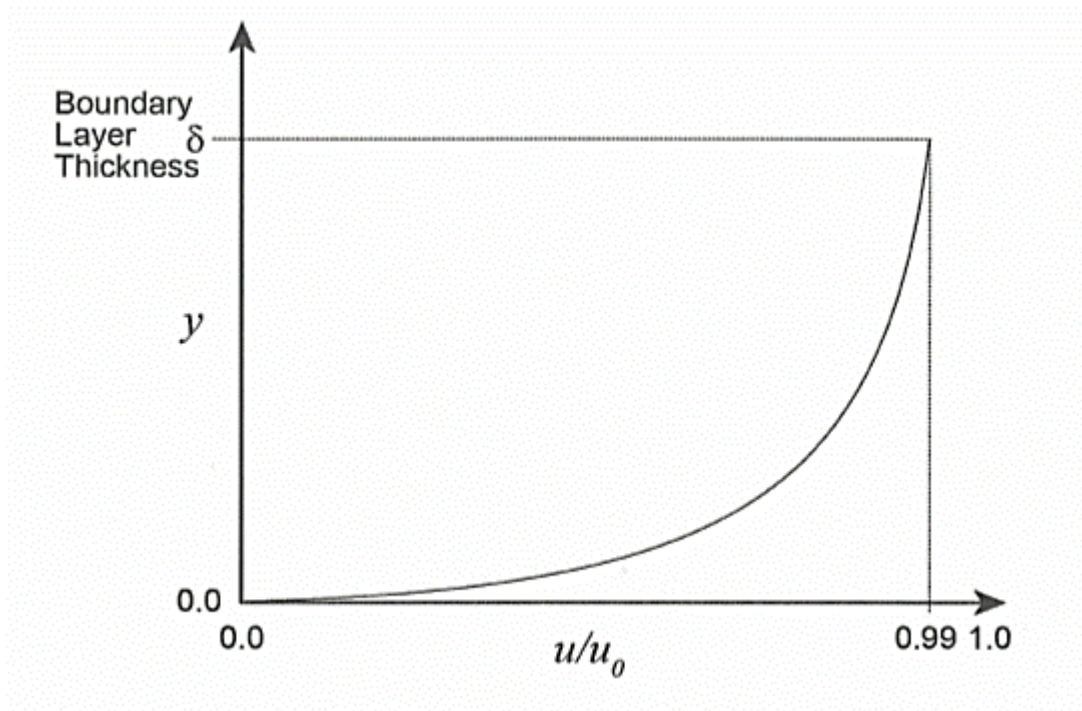


Figure 5: Velocity in the Boundary Layer

From this, at any horizontal position x the boundary layer thickness can be found simply by measuring the distance from the surface where the local measured velocity (u) equals the free stream velocity (u_0).

As a good comparison against the theory, for any horizontal position x , a plot of actual vertical position y of the measuring point against actual u/u_0 will produce a similar curve with a maximum u/u_0 value of approximately 1 (0.99 in textbooks). At this point, the distance on the y axis should give an approximate value for the boundary layer thickness (δ).

Note: In textbooks, it may be seen that the boundary layer is defined as the where the local velocity = 99% of free stream velocity ($u/u_0 = 0.99$).

Figure 6: Finding Boundary Layer Thickness at any Position x

Distance x and Boundary Layer Growth

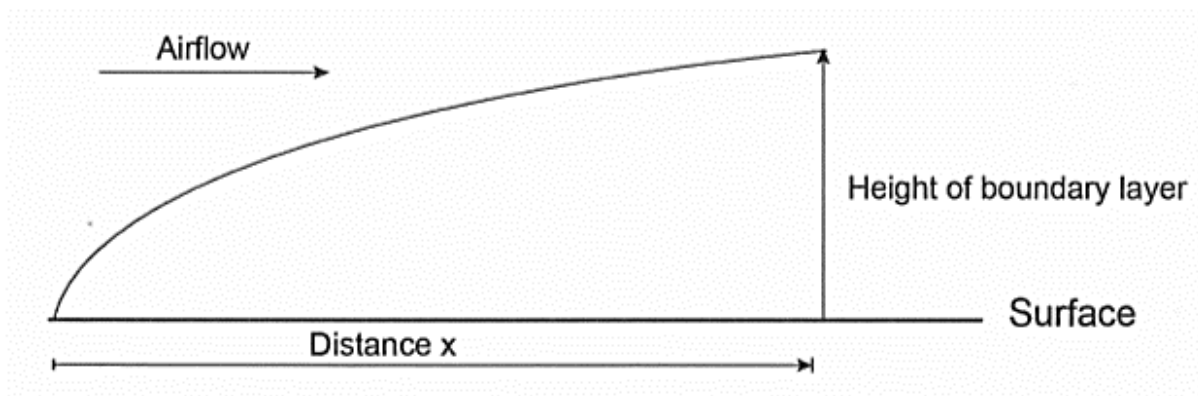


Figure 7: Distance and Boundary Layer

As the air moves along the surface, the boundary layer thickness increases with distance x . Several variables affect this boundary layer growth:

- Air conditions - free stream velocity and Reynolds number (turbulent or laminar flow)
- The sharpness of the leading edge - affects the initial boundary thickness
- The surface finish - rough or smooth

Generally, higher velocities give a thicker boundary layer, a sharp leading edge gives small initial boundary thickness compared to a blunt leading edge and a smooth surface produces a smaller

thickness. In practice, the air properties (and hence Reynolds number) are the biggest influence on the thickness.

Due to this fact, the approximate boundary thickness can be predicted based on the Reynolds number and the distance x along the surface for any given surface finish - in this case a smooth flat plate.

For ***laminar*** flow boundary layer thickness:

$$\delta \approx \frac{4.91x}{\sqrt{Re_x}} \quad (8)$$

For ***turbulent*** flow boundary layer thickness:

$$\delta \approx \frac{0.382x}{Re_x^{0.2}} \quad (9)$$

Notes

You will be running the experiment and collecting data points to use in the comparison. Use this section to record any notes you have.

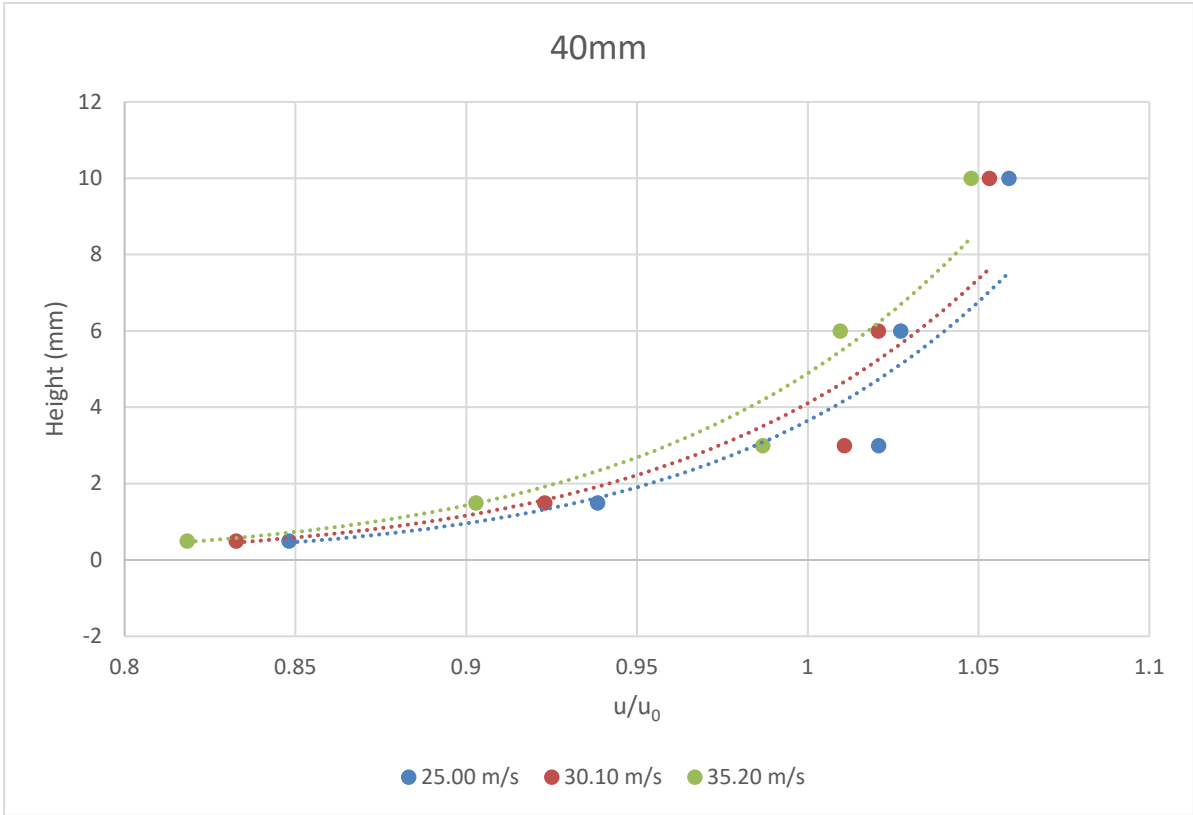
Results:

Use the format shown here. For each of the three test velocities, provide the following for each of the aerofoil rake (5 rakes).

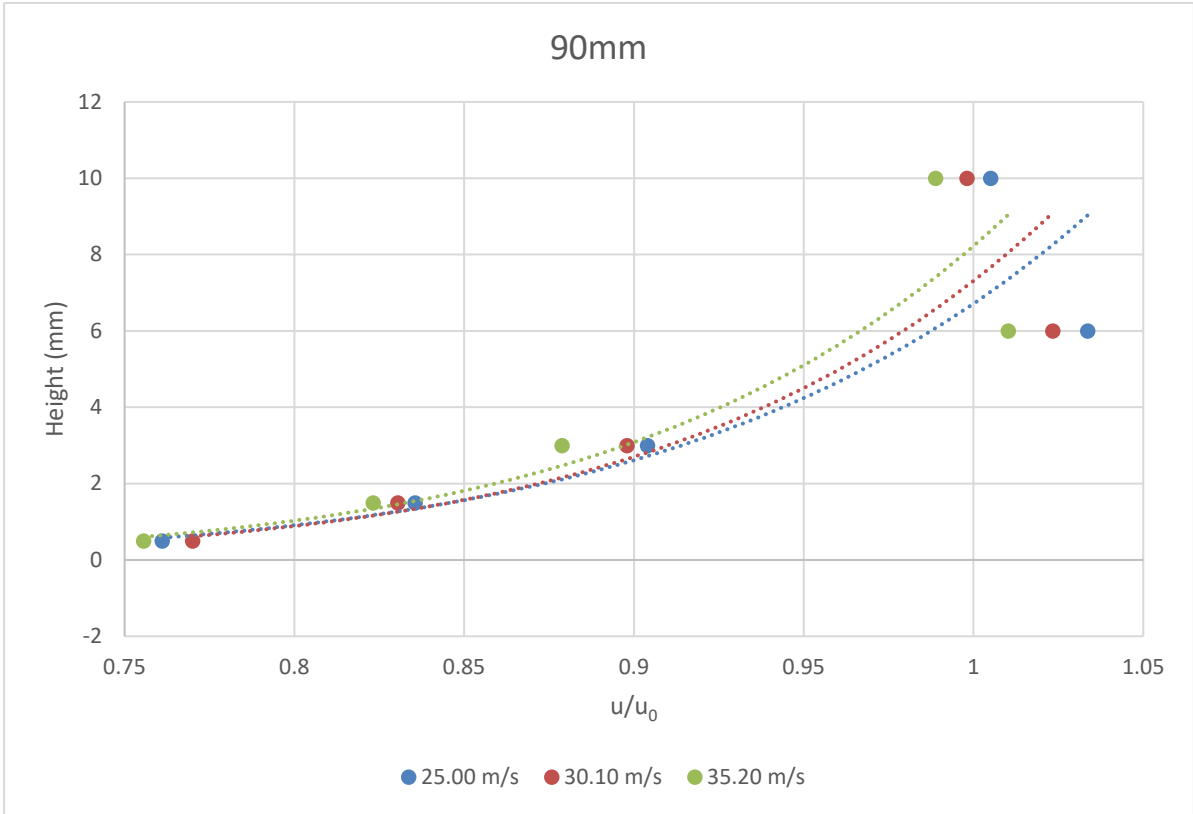
- A single graph at each measurement location, with all three velocities overlaid on the same axes. Ensure appropriate trend lines are used, and make sure to use a consistent colour scheme to differentiate each velocity.
- A table displaying your data of the measured boundary layer height and the theoretical boundary layer height at each measurement location.
- One graph showing boundary layer growth over the flat plate. Plot both the theoretical and experimental boundary layer height at each measurement location for all three velocities. Plot measured heights as data points, and theoretical displacement thickness as a curve. See example below for reference.

Examples are shown on the next page, note that this data is fictional so should not be used to check your results, it is merely there as a guide and should be deleted. Refer to the rubric posted on canvas for further details.

Note: Submit exactly 6 graphs total (5 velocity profile overlays—one per measurement location + 1 boundary layer growth overlay). Use the example table format as a guide (delete fictional data). Ensure graphs are legible

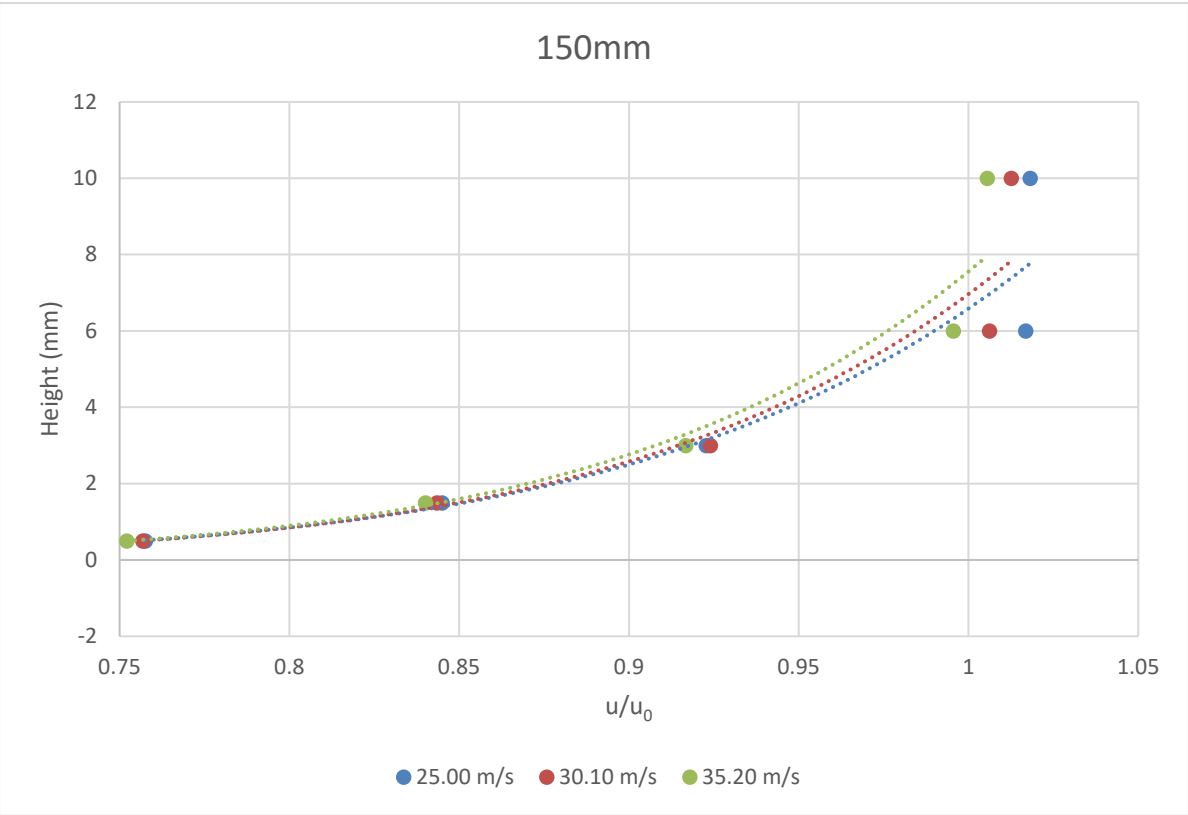


Freestream velocity 40mm: m/s	25.00	30.10	35.20
Measured Thickness δ_M (mm)	2.44	2.65	3.41
Theoretical Thickness δ_T (mm)	2.96	2.70	2.50



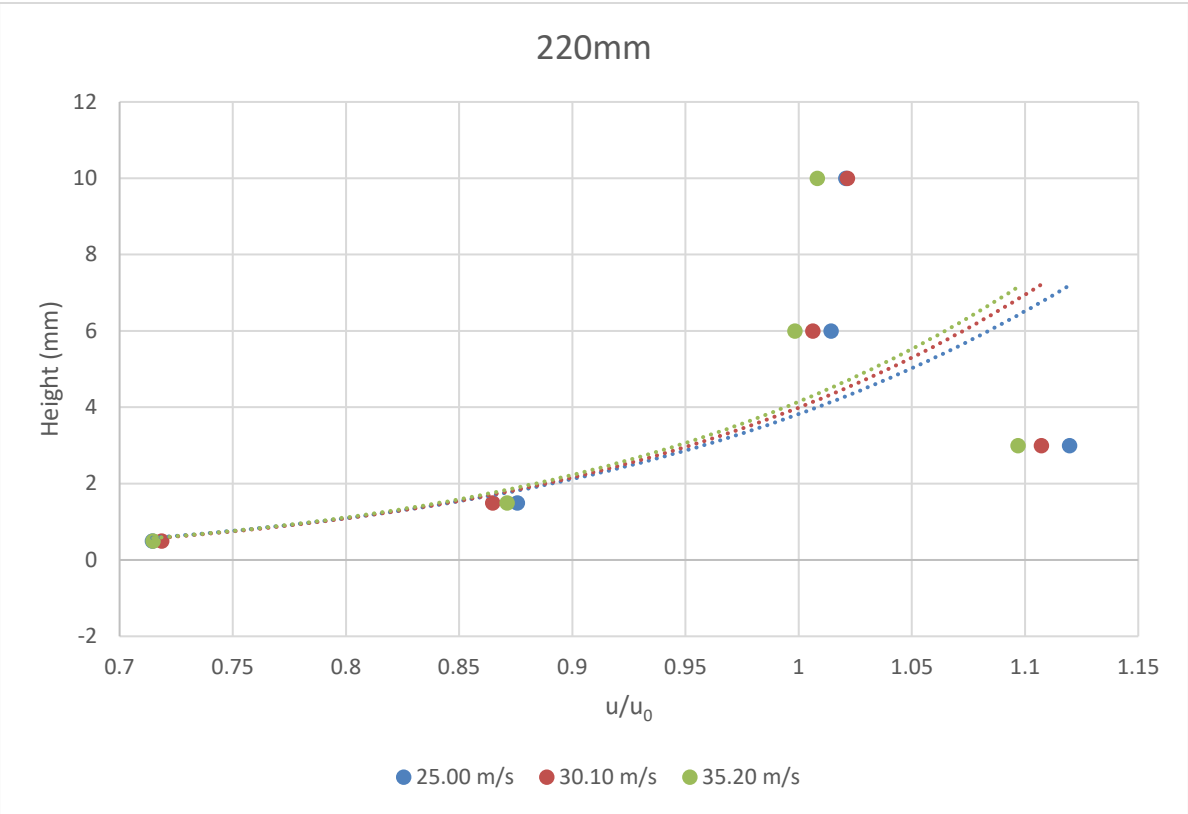
Freestream velocity 90mm: m/s	25.00	30.10	35.20
Measured Thickness δ_M (mm)	4.99	5.20	5.54
Theoretical Thickness δ_T (mm)	5.13	4.68	4.33

150mm



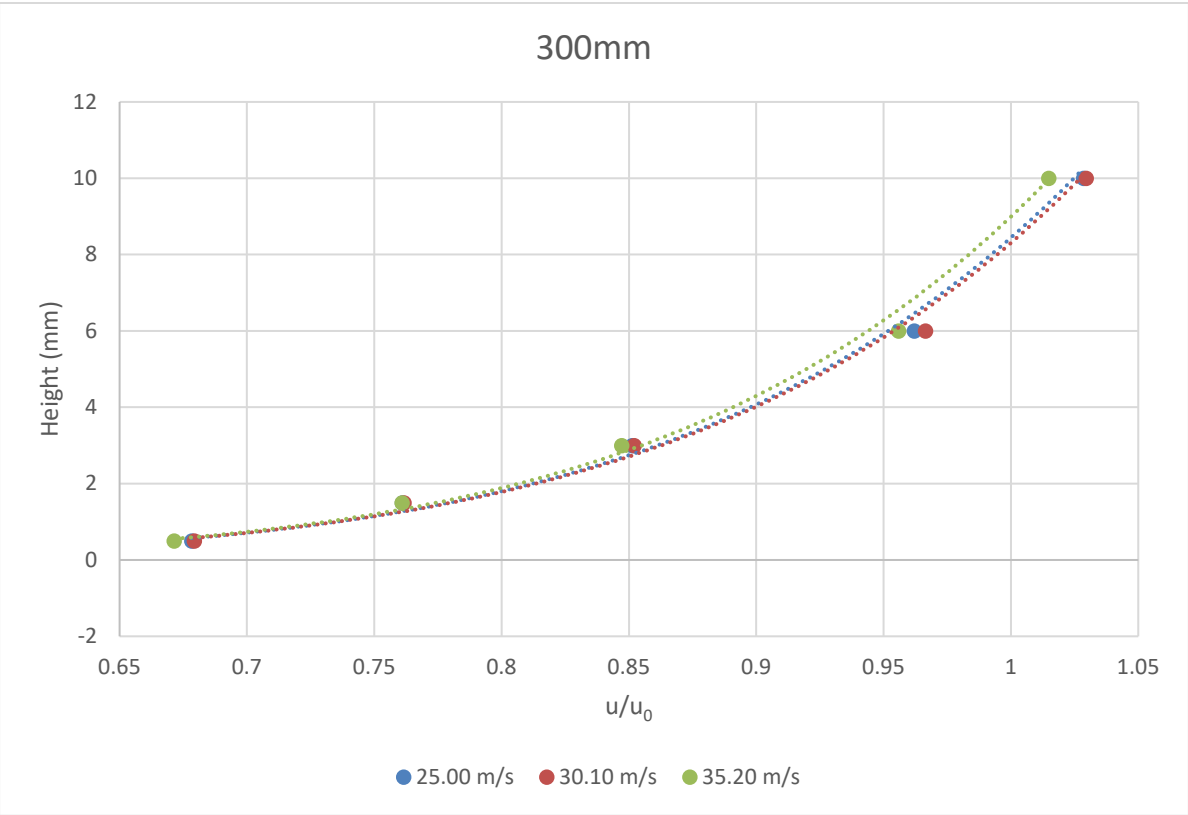
Freestream velocity 150mm: m/s	25.00	30.10	35.20
Measured Thickness δ_M (mm)	5.14	5.41	5.78
Theoretical Thickness δ_T (mm)	7.26	6.62	6.12

220mm

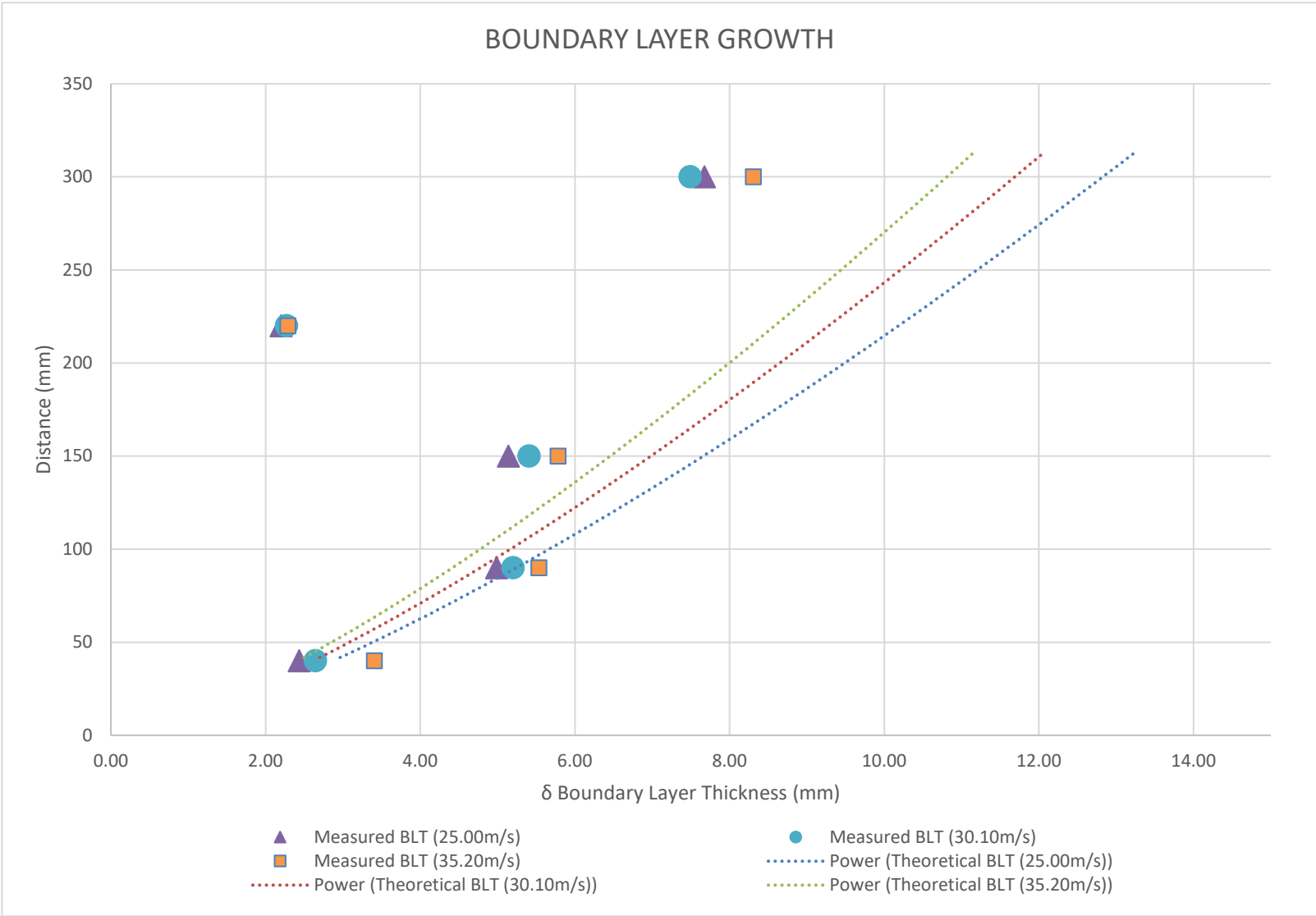


Freestream velocity 220mm: m/s	25.00	30.10	35.20
Measured Thickness δ_M (mm)	2.20	2.27	2.29
Theoretical Thickness δ_T (mm)	10.27	9.36	8.65

300mm



Freestream velocity 300mm: m/s	25.00	30.10	35.20
Measured Thickness δ_M (mm)	7.68	7.49	8.31
Theoretical Thickness δ_T (mm)	13.26	12.08	11.17



y (mm)	Theoretical BLT (25.00m/s)	Theoretical BLT (30.10m/s)	Theoretical BLT (35.20m/s)	Measured BLT (25.00m/s)	Measured BLT (30.10m/s)	Measured BLT (35.20m/s)
0.5	2.9641	2.7012	2.4980	2.4379	2.6456	3.4078
1.5	5.1339	4.6785	4.3266	4.9883	5.2033	5.5365
3	7.2605	6.6165	6.1188	5.1402	5.4086	5.7846
6	10.2678	9.3571	8.6532	2.2022	2.2746	2.2887
10	13.2557	12.0800	11.1713	7.6758	7.4941	8.3086

1. In two dot points describe any patterns that you see

- With all five velocity ratio vs tap height, it shows a distinct exponentially height increasing curve. This suggests that the greater the y height, the quicker the local velocity travels relative to the freestream velocity. This may have occurred as a result of the boundary layer effect where viscous forces force the air velocity to be lower at closer points to the plate.
- Another pattern which can be seen is an association between the increase in freestream velocity and the measured boundary layer thickness. For all airfoils, the measured boundary layer thickness was greatest when the freestream velocity was 35.20 m/s, and in four of the five airfoils the boundary layer thickness was second greatest for a 30.10 m/s freestream velocity.

2. In two dot points describe any anomalies that you see

- For the measured boundary layer thickness, for the airfoils sitting 220 mm from the front of the plate they record a vast decrease in the boundary layer thickness than is expected from the theoretical. This is for all airspeeds, the thickness drops to just above 2mm, where from the theoretical we would expect it to be at 8-10mm. This is likely a result from a faulty tap 18, it is vastly different from the other taps outputting a positive number.
- Another anomaly is with the 220mm velocity profile where a 5mm height where the velocity ratio spikes relative to the trendline. This is consistent with each windspeed tested. These datapoints were recorded from tap 18, suggesting it again was due to an issue with the tap outputting the wrong data.

3. *If this experiment were to be repeated, what changes would you implement to enhance the accuracy and reliability of the results?*

Initially I would run pre-emptive tests to ensure all sources of data collection was operating to standard to ensure the most accurate results. This wasn't done during these tests as can be seen from the data output in tap 18 which output significantly incorrect data which has had a major impact in the results as can be seen from both the boundary layer growth and velocity ratio v height graphs.

Another change I would implement would be to employ a more accurate setting of windspeed. Using the current method of a student attempting to get as close to the target speed as possible is unreliable and produces outputs that differ slightly to the brief. While the measurements use the correct speeds and therefore has no effect on the actual data, it leads to inconsistent data points at which data can be compared to. To fix this the lab would need to employ some new technology where the windspeed can be automatically set.

A better method to get the temperature and pressure data should also be used. For this lab it was gained by reading a physical thermometer and a pressure gauge. These however are difficult to obtain absolute data values as human or parallax errors may occur. While steps were taken to minimise these, they still occur and have slight effects on the results. A way to diminish these effects is to use a digital pressure sensor and pressure gauge, disallowing chances for human errors to occur.

4. *Using the measured data, calculate the drag on the bottom surface of the geometry, assuming it is a perfectly flat 3D plate while accounting for the laminar-to-turbulent transition in the 20 m/s test run. Show your calculations*

Note: no data test at 20 m/s so went with next lowest at 25.00 m/s

$$u_0 = 25.00, \quad \rho = 1.21, \quad \mu = 18.22 \times 10^{-6}, \quad L = 0.526, \quad W = 0.300, \quad v = 15.06 \times 10^{-6}$$

$$Re_L = \frac{\rho U_0 L}{\mu} = \frac{1.21 \times 25.00 \times 0.526}{18.22 \times 10^{-6}} = 873298.57$$

$\therefore$ Transition Occurs

$$x_{cr} = \frac{Re_{cr} v}{u_0} = \frac{(5 \times 10^5) \times (15.06 \times 10^{-6})}{25.00} = 0.3012 \text{ m}$$

$$A = Re_{cr} \times \left(\frac{0.074}{Re_{cr}^{0.2}} - \frac{1.328}{\sqrt{Re_{cr}}} \right) = (5 \times 10^5) \times \left(\frac{0.074}{(5 \times 10^5)^{0.2}} - \frac{1.328}{\sqrt{(5 \times 10^5)}} \right) = 1742.65$$

$$C_f = \frac{0.074}{Re_L^{0.2}} - \frac{A}{Re_L} = \frac{0.074}{873298.57^{0.2}} - \frac{1742.65}{873298.57} = 0.002802$$

$$q = \frac{1}{2} \rho u_0^2 = \frac{1}{2} \times 1.21 \times 25.00^2 = 378.125 \text{ Pa}$$

$$A_{pl} = L \times W = 0.526 \times 0.300 = 0.1578 \text{ m}^2$$

$$D = C_f q A_{pl} = 0.002802 \times 378.125 \times 0.1578 = 0.167 \text{ N}$$

References

Editor Engineeringtoolbox 2026, "Air viscosity: dynamic and kinematic viscosity at various temperatures and pressures," viewed <https://www.engineeringtoolbox.com/air-absolute-kinematic-viscosity-d_601.html?vA=22&units=C#>.